

Mini Report

Problem Statement

Insurance policies are long, complex, and full of jargon. Comparing multiple policies manually is time-consuming and confusing. Users need a tool that summarizes, extracts key metrics, and visualizes comparisons clearly.

Why We Chose This Problem

Policy comparison is a real-world pain point for customers and agents. Most people struggle to evaluate which insurance plan is better for them. Automating this with AI saves time and improves decision-making.

How GenAI Helped

We used **Google Gemini** to:

- Summarize long policy documents into structured paragraphs.
- Extract key metrics (Sum Insured, Waiting Period, Claim Settlement Ratio).
- Fill gaps intelligently when explicit values were missing.
- Enable a conversational **Q&A agent** to answer user queries about policies.

GenAI allowed us to handle unstructured text flexibly without building rule-based NLP pipelines from scratch.

Key Learnings

1. **Prompt engineering is iterative** - first outputs were inconsistent until we constrained style (JSON, paragraph).
2. **Regex + AI hybrid approach** - works best - regex catches clear patterns, AI fills missing values.
3. **User experience matters** - adding tables, dropdowns, and visualizations made results far more usable.
4. **Structured outputs are essential** - forcing JSON ensured clean extraction.